

Majme singulárny rozklad matice A ako v predchádzajúcej úlohe. Teda $A = VDU^T$ kde

$$\begin{aligned} v_1 &= (1, 1, 1)^T & \sigma_1 &= \sqrt{3} & u_1 &= \sigma_1^{-1} A^T v_1 \\ v_2 &= (1, 0, -1)^T & \sigma_2 &= 3 & u_2 &= \sigma_2^{-1} A^T v_2 \\ v_3 &= (1, -2, 1)^T & \sigma_3 &= 3 & u_3 &= \sigma_3^{-1} A^T v_3 \end{aligned}$$

$$u_4 = u_1 \times u_2 \times u_3$$

$$U = (u_1, \ u_2, \ u_3, \ u_4) \tag{1}$$

$$V = (v_1, \ v_2, \ v_3) \tag{2}$$

$$D = \begin{pmatrix} \sqrt{3} & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{pmatrix} \tag{3}$$

Zvoľme

$$E = \begin{pmatrix} (\sqrt{3})^{-1} & 0 & 0 \\ 0 & 3^{-1} & 0 \\ 0 & 0 & 3^{-1} \\ 0 & 0 & 0 \end{pmatrix} \tag{4}$$

Matica E je z prava inverzná k D a teda $DE = I_3$. Z toho ľahko dostávame že $EDE = E$, $DED = D$ a $DE = (DE)^T$. Výpočtom tiež ľahko dostaneme že $ED = (ED)^T$. Ďalej definujme $B = UEV^T$, potom

$$ABA = VDU^T UEV^T VDU^T = VDEDU^T = VDU^T = A \tag{5}$$

$$BAB = UEV^T VDU^T UEV^T = UEDEV^T = UEV^T = B \tag{6}$$

ďalej

$$AB = VDU^T UEV^T = VDEV^T = (VDEV)^T = (AB)^T \tag{7}$$

$$BA = UEV^T VDU^T = UEDU^T = (UEDU^T)^T = (BA)^T \tag{8}$$

Teda B má všetky požadované vlastnosti. $\square$